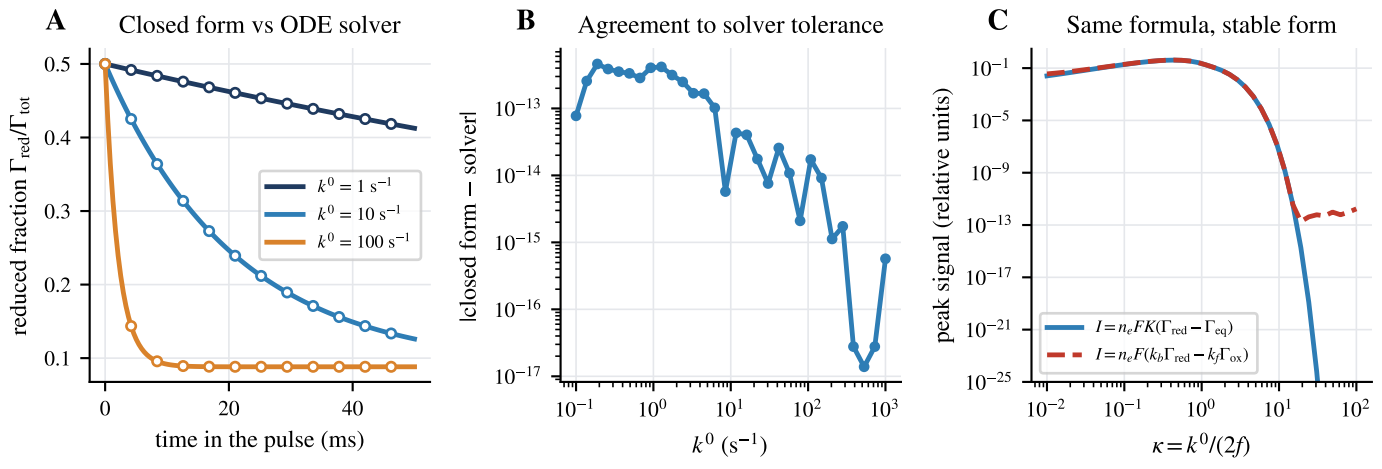
A
Closed form vs ODE solver
reduced fraction $\Gamma_{red}/\Gamma_{tot}$
time in the pulse (ms)
$k^0 = 1\ s^{-1}$
$k^0 = 10\ s^{-1}$
$k^0 = 100\ s^{-1}$
B
Agreement to solver tolerance
|closed form − solver|
$k^0\ (s^{-1})$
C
Same formula, stable form
peak signal (relative units)
$\kappa = k^0/(2f)$
$I = n_e F K(\Gamma_{red} - \Gamma_{eq})$
$I = n_e F(k_b\Gamma_{red} - k_f\Gamma_{ox})$